

Transformations and vectors

Igcse Mathematics

This handout covers Topic 7, Transformations and vectors. Parts marked **(Extended)** are only tested on the Extended papers; everything else is for both levels. Vectors as a whole topic are Extended.

Transformations

A **transformation** 变换 changes the position or size of a shape. The original is the *object* and the result is the *image*. There are four types. When asked to **describe** one, you must name the type and give all the details below.

Reflection

A **reflection** 反射 flips the shape over a **mirror line** 对称轴. Each image point is the same distance from the line as the object point, on the other side.

- To describe it, give the **equation of the mirror line** (for Core, a horizontal or vertical line; Extended allows any line such as $y = x$).

Worked example. Reflect the point $(3, 2)$ in the y -axis. Only the sign of x changes: the image is $(-3, 2)$. (In the x -axis it would be $(3, -2)$.)

Rotation

A **rotation** 旋转 turns the shape about a fixed point, the **centre** 中心 of rotation, through multiples of 90° .

- To describe it, give the centre, the angle, and the direction (clockwise or anticlockwise).

Worked example. Rotate $(3, 1)$ by 90° anticlockwise about the origin. The rule is $(x, y) \rightarrow (-y, x)$, so the image is $(-1, 3)$.

Enlargement

An **enlargement** 放大 changes the size by a **scale factor** 比例因子 k , measured from a fixed centre. Each distance from the centre is multiplied by k .

- To describe it, give the centre and the scale factor. For Extended, k may be **fractional** (the shape gets smaller) or **negative** (the image appears on the other side of the centre).

Worked example. Enlarge $(1, 2)$ from the origin by scale factor 2. Multiply both coordinates: the image is $(2, 4)$.

Translation

A **translation** 平移 slides the shape with no turning, by a **vector** 向量 written as a **column vector** 列向量 $\begin{pmatrix} x \\ y \end{pmatrix}$ (x across, y up).

Worked example. Translate $(5, 3)$ by $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$: move 2 left and 4 up to get $(3, 7)$.

(**Extended:** a question may ask you to combine two transformations and describe the single transformation that has the same effect.)

Vectors in two dimensions (Extended)

A vector has both size and direction. It can be written as a column vector, as $\overrightarrow{AB}$ (from A to B), or in bold as $\mathbf{a}$.

- **Add or subtract** by working on the top and bottom numbers separately.
- **Multiply by a scalar** 标量 (an ordinary number) by multiplying both numbers.

Worked example. If $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$, then

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \quad 3\mathbf{a} = \begin{pmatrix} 9 \\ 3 \end{pmatrix}.$$

Magnitude of a vector (Extended)

The **magnitude** 模 (length) of a vector is found with Pythagoras. For $\begin{pmatrix} x \\ y \end{pmatrix}$,

$$\left| \begin{pmatrix} x \\ y \end{pmatrix} \right| = \sqrt{x^2 + y^2}.$$

Worked example. The magnitude of $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

Vector geometry (Extended)

A vector can be drawn as a **directed line segment** 有向线段 (an arrow). The **position vector** 位置向量 of a point is the vector from the origin O to that point.

A key idea: the vector from A to B is

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a},$$

where $\mathbf{a}$ and $\mathbf{b}$ are the position vectors of A and B .

Two vectors are **parallel** 平行 if one is a scalar multiple of the other (for example $\overrightarrow{AB} = 2\overrightarrow{CD}$). Three points are **collinear** 共线 (in a straight line) if the vectors between them are parallel and share a point. You can express any vector in terms of two **coplanar** 共面 vectors.

Worked example. O is the origin, with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. M is the midpoint of AB . Find $\overrightarrow{OM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

$$\overrightarrow{OM} = \mathbf{a} + \frac{1}{2}\overrightarrow{AB} = \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{b}).$$